

Circuit Theory - Question Bank

Course: Circuit Theory | B.Tech ECE | Semester 1 | Year 1

1. According to Ohm's law, V is equal to:

- a) I / R
- b) $I \times R$
- c) R / I
- d) $I + R$

Answer: b

2. What is the SI unit of resistance?

- a) Volt
- b) Ampere
- c) Ohm
- d) Watt

Answer: c

3. Kirchhoff's current law states that:

- a) the sum of voltages in a loop is zero
- b) the sum of currents at a node is zero
- c) power is conserved
- d) energy is conserved

Answer: b

4. In a series circuit, the current is:

- a) the same through all elements
- b) divided among elements
- c) always zero
- d) always alternating

Answer: a

5. What is the SI unit of capacitance?

- a) Henry
- b) Farad
- c) Ohm
- d) Siemens

Answer: b

6. Power dissipated in a resistor can be calculated as:

- a) $V \times I$
- b) $I^2 \times R$
- c) V^2 / R
- d) all of these

Answer: d

7. Thevenin's theorem simplifies a network to:

- a) one voltage source and one resistor
- b) two voltage sources
- c) one current source
- d) a single capacitor

Answer: a

8. What is the SI unit of frequency?

- a) Second
- b) Hertz
- c) Joule
- d) Tesla

Answer: b

9. The inductive reactance X_L is equal to:

- a) $2 \times \pi \times f \times L$
- b) $1 / (2 \times \pi \times f \times C)$
- c) R
- d) L / C

Answer: a

10. In a DC circuit, a capacitor behaves like:

- a) a short circuit
- b) an open circuit
- c) a resistor
- d) an inductor

Answer: b

11. State and explain Kirchhoff's voltage law.

(Write your answer in the space provided.)

12. Explain the main differences between AC and DC.

(Write your answer in the space provided.)